

D TRACK

“Track the unseen. Stop the chain.”



TEAM NAME : ZERO_CIVIC_SENSE

Our Team

Zero_Civic_Sense



VEDANT

DTU CSE 2028

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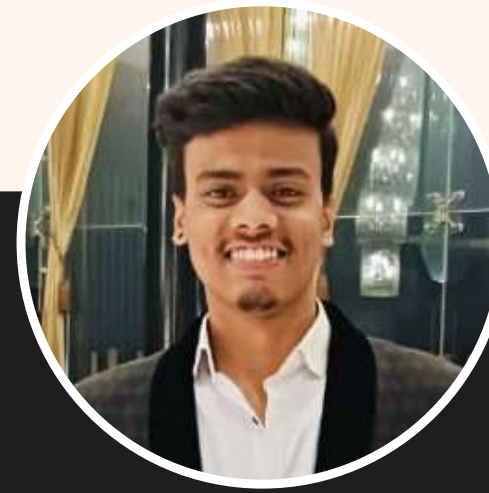


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YASH

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PROBLEM STATEMENT

- Drug trafficking has gone **Digital & Invisible** and our detection tools are a decade behind
- Traffickers **don't say illegal words** anymore.

Details:



In its 2025 Annual Report, the International Narcotics Control Board:

- India recorded over **1 lakh** drug-related cases in 2025 alone — and that's only what was detected
- **Telegram** alone hosts thousands of active **drug distribution channels** — many operating openly with emoji-coded menus
- Blockchain analytics firm Chainalysis estimates **\$24.2 billion in crypto** was received by illicit addresses in 2023



The Silo Problem: Drug networks no longer operate in alleys. They advertise on **Instagram** using coded emoji, transact via **Telegram bots**, and collect payment in **crypto** across three platforms, three identities, zero obvious connection.



The Language Problem: Traffickers use coded language (emojis, slang, patterns) instead of keywords, bypassing **traditional detection systems**.



The Scale Problem: Investigators face massive data with limited resources; without automation, major networks remain hidden while only small cases get caught.



PROPOSED SOLUTION

AI-Powered Trafficking Detection System



CORE FEATURES:

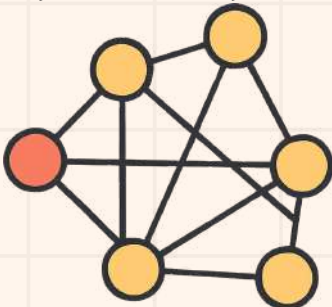
Slang-to-Signal Detection Engine

- **Problem it solves:** Traffickers don't say illegal words anymore
- Instead of looking for specific words, **this engine looks for patterns of intent.** It reads a post the way a trained investigator would noticing that
- It **converts every post into a mathematical "meaning vector" using a machine learning model** (sentence-transformers). It then compares that vector against a reference database of confirmed trafficking patterns. If the meaning is close enough flags it



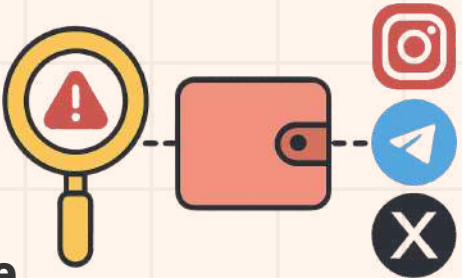
Shadow-Graph Visualizer

- **Problem it solves:** Linked accounts alone don't reveal network structure or key players.
- Stores identities and wallets in a graph database :**
- Nodes = accounts
 - Edges = shared wallet, links, or phone
- Renders an interactive network map** to explore connections, clusters, and money flows.
- Reveals:**
- Network scale (isolated vs organized)
 - Central wallets (collection points)
 - "Clean" accounts promoting operations



Cross-Platform Handle Stitching

- **Problem it solves:** Traffickers use different platforms (e.g., clean Instagram, active Telegram), making identities hard to link.
- **Uses crypto wallet address as a unique fingerprint** (public, consistent across platforms)
Matches profiles across Instagram, Telegram, and X using
 - Same wallet address
 - Phone number / bio links
 - Similar username patterns



Links **matched accounts into a single unified identity profile.**

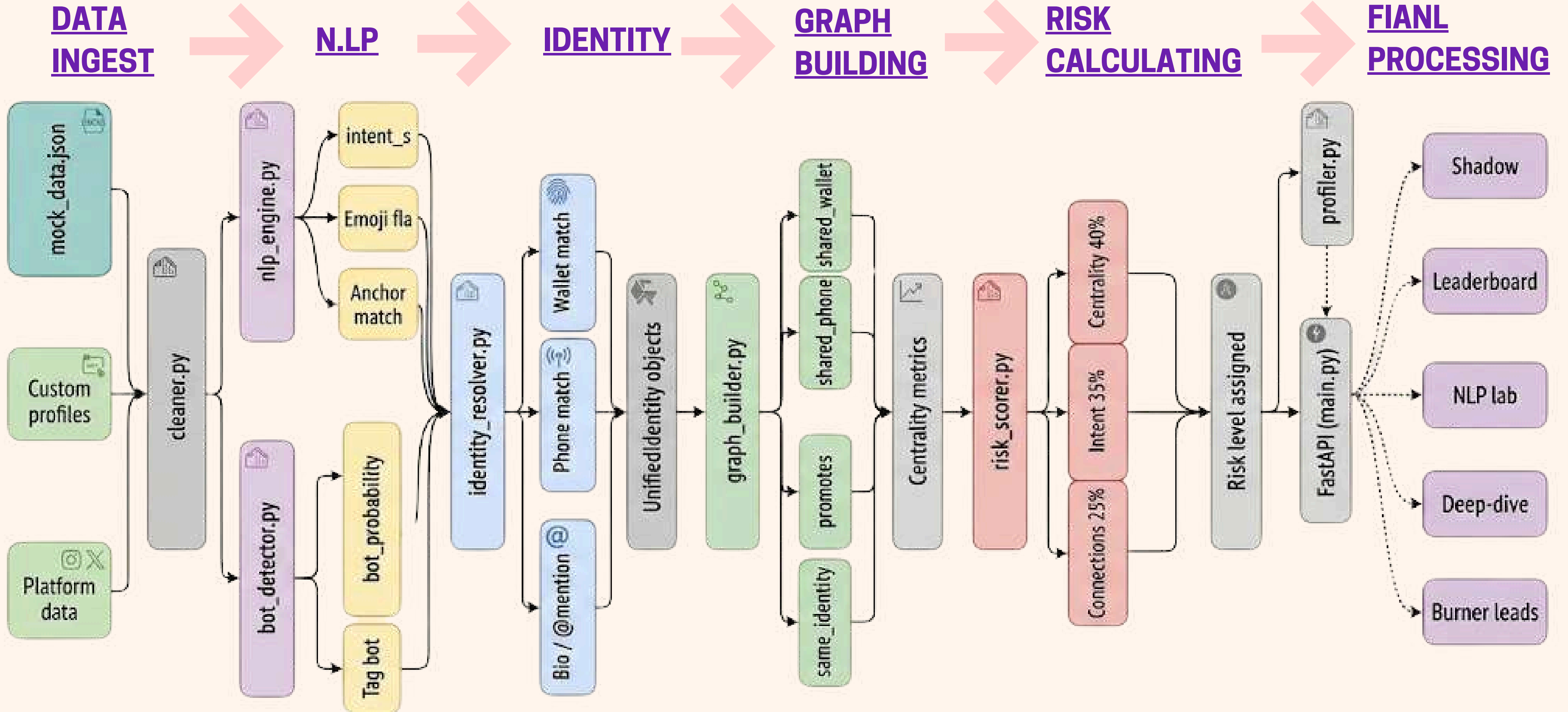
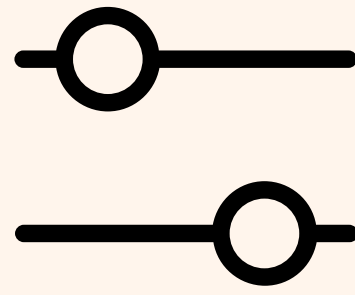
Risk Scoring Dashboard

- **Problem it solves:** Investigators **can't manually review** hundreds of flagged accounts.
- Assigns each identity a Risk Score (0-100)** based on:
- Graph Centrality: role in network (hub vs peripheral)
 - Blockchain Volume: crypto transaction activity
 - Activity Frequency: posting/operational intensity
- Ranks accounts so high-risk targets appear first.**



COMPLETE RISK

ASSESSMENT PIPELINE



TECHNICAL ARCHITECTURE

LAYER BASED APPROACH

DATA INGESTION

Instagram / X Scraper
Public posts ,bios ,geostag
Using: Playright , Apify

Telegram Collector
Channel ,bots , media
Using: Telethon

BlockChain API
Wallet history , Volumes
Using: Etherscan , Solscan

PROCESSING

Slang-to-Signal Engine
Converts every scraped post into a semantic embedding vector

OCR Image Pipeline
Extract text from images

Geo Extractor
Location mentions, exif tags

IDENTITY RESOLUTION

Handle Sticing Engine
Public posts ,bios ,geostag
Using: Playright , Apify

Entity Deduplication
Merge duplicate identities
Using: MinHash LSH

SCORING & VISUALISATION

Risk Scoring engine
0-100 per identity, multi-signal
Using: Python · Chain APIs

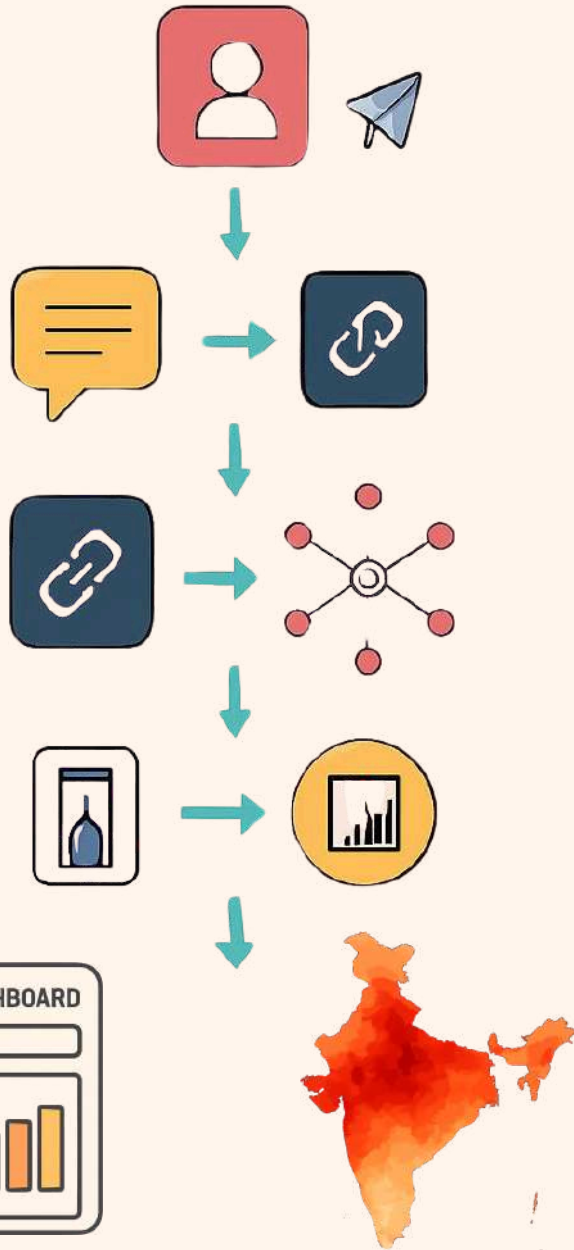
Graph Database
Nodes,edges, centrality scoring
Using: Neo4j GDS

Shadow-Graph Visualizer
Interactive network map, exportable

RISK SCORING DASHBOARD
Ranked targets, filterable

Geospatial Heatmap
Hot-zone mapping, clusters

DATA FLOW



TECH STACK:



WHY D-TRACK WINS: Features



Robust False Positive Management

Employs a 0.72 confidence threshold and mandatory corroboration (ML intent + hard signal like a wallet/handle) to eliminate false positives, backed by an active learning loop that trains on analyst corrections.



Adversarial Sanitization:

Neutralizes evasion tactics by normalizing Unicode "lookalikes" and using perceptual hashing (pHash) to catch modified images, ensuring only clean, deduplicated data hits the model.



Persistent Entity Resolution:

Defeats burner accounts by tracking Soft Fingerprints—linking rotated identities through writing style (stylometry), operational habits, and crypto co-spending clusters.



Ethical Guardrails:

Prioritizes legal compliance via a 90-day raw data purge and a "Human-in-the-Loop" design that focuses automated power on the 90% of traffickers who lack perfect operational security.

BUSINESS APPROACH



State Cybercrime Units (Primary Focus)



The Gap:

- Units are **overwhelmed by the sheer volume of internet crimes** and lack tools to analyze complex digital footprints

Our Solution:

- Automated Identity Stitching: Our "Shadow-Graph" **reduces manual investigative time from 10 days to 10 seconds** by instantly linking identities across platforms using public wallet fingerprints.

Revenue Model:

- Annual Maintenance Contract (AMC) & License Fee.
- Instead of complex per-user pricing, we charge a flat annual fee per state unit or district headquarters.

Fintech Platforms (Recurring Revenue)



The Gap:

- Current systems flag suspicious transaction volumes after they happen**, but miss the "intent" happening on social media.

Our Solution:

- A plug-and-play risk-scoring engine that scans content and wallet addresses at the point of interaction.
- Pattern-of-Intent Detection: Instead of just looking for "Illegal drugs," we flag mathematical

Revenue Model:

- "API Volume + Seat" Pricing: * API Tier: Tiered pricing based on the number of transactions scanned.
- Dashboard Access: ₹15,000 – ₹40,000/month per analyst seat.

Research & NGOs (Grant & Data Partnership)



The Gap:

- These organizations have **global stats but lack granular**, real-time data on how digital trafficking routes shift across the Indian Ocean and South Asia.

Our Solution:

- Real-time Intelligence Feed: A dashboard that provides live trends on new synthetic drug names and digital transit routes being used in the region.

Revenue Model:

- Grant-Funding & Indirect Monetization
- Research Grants: Applying for "Cyber Security Innovation" grants from MeitY or international bodies like the UNODC

IMPACT & BENEFITS



10× Faster Investigations:
Reduces weeks of manual cross-referencing to under 10 minutes — investigators spend time acting on leads, not chasing them.



Cross-Platform Identity Resolution:
Stitches aliases across Instagram, Telegram, and X using shared wallet addresses and bio-link patterns into a single unified profile.



The Anonymity Gap — Closed:
Crypto wallet acts as a permanent fingerprint. Even with aliases and coded language, D-TRACK links the "clean" persona to the distribution network.



Scalable & Broadly Applicable:
Platform-agnostic architecture extends to TikTok, WhatsApp, and beyond. Same logic applies to counterfeit goods, financial fraud, and human trafficking networks.

FROM SCATTERED SIGNALS TO PROSECUTABLE INTELLIGENCE

Metric	Impact
Investigation Time	Reduced from weeks → 10 minutes of automated graph traversal
Identity Resolution	3+ platforms unified under one wallet fingerprint
Deployment Reach	Integrates with CCTNS & INTERPOL I-24/7 workflows
Data Privacy	100% public data — no private surveillance involved